

Gemini 2.5 Flash · Free Tier Analysis

Why Atomic Ops Agent Hits Rate Limits

Atomic Group · Supply Chain Dashboard · May 2026

The Problem in One Number

1 brief = 5 API calls

Not one. **Five.**

Gemini Free tier allows **10 requests per minute.**

Refresh the brief twice — you've burned **all 10.**

Why 1 Brief = 5 Calls

The Tool-Call Loop

```
Round 1 → Gemini: "I need query_pos"  
Round 2 → Gemini: "Now query_purchase_decisions"  
Round 3 → Gemini: "Now query_batches"  
Round 4 → Gemini: "Now overdue POs"  
Round 5 → Gemini: writes the final brief text
```

Each arrow = **one billable API request**.

The model can't see tool results until we send them back in the next round.

Tokens Snowball Each Round

Conversation Replays Every Time

LLMs are **stateless**. Every round resends the entire history.

ROUND	WHAT'S SENT	TOKENS
1	system + prompt	2K
2	+ tool_result_1	6K
3	+ tool_result_2	12K
4	+ tool_result_3	20K
5	+ tool_result_4	28K
Total per brief		~68K

Free tier TPM = 250K. **3 briefs/min hits the cap.**

Tool Results Are Heavy

Each result $\approx$ 4–6K tokens

query_purchase_decisions returns 30 SKUs $\times$ 15 fields each:

```
sku, brand, product_name, available, l3m_avg,  
stock_months, status, safety, moq, lead_time_days,  
suggest_qty, unit_cost, amount, supplier_name,  
payment_terms, active_po
```

The agent only needs the **top 5** for the brief.

We're sending **6 $\times$ more data** than necessary.

Hidden Overhead

System Prompt Repeats 5×

**800**

BUILDSYSTEMPROMPT TOKENS

**600**

DAILYBRIEFSEED TOKENS

**7,000**

RESENT PER BRIEF

Both prompts are resent **every round** — that's **7,000 tokens of pure boilerplate** per brief, before any actual data.

The Testing Trap

Role Switching Multiplies Everything

```
useEffect(() => {  
  if (briefedFor === current.name) return  
  runChat(seedPrompt, { isBrief: true }) // auto-fires  
}, [current.name])
```

Viewing as Syuen → Jun Ye → Grace in quick succession:

3 briefs × 5 calls = 15 requests in seconds → instant 429.

The Math

Free Tier vs. Our Usage

ACTION	REQUESTS	TOKENS
1 brief	5	~68K
Switch 3 roles	15	~200K
Free tier limit	10/min • 250/day	250K/min

Verdict: **Architecture is too heavy for Free tier** at current testing/usage patterns.

Recommended Optimizations

4 Fixes (No Paid Plan Needed)

- 1. **Slim tool results** — `slice(0, 5)` instead of `(0, 30)` → save ~30K tokens/brief
- 2. **Parallel tool calls** — collapse 5 rounds into 2
- 3. **Cache briefs** — reuse within a 10-minute window
- 4. **Manual role-switch brief** — no auto-fire on viewer change

Combined effect: **~80% reduction** in daily requests.
Free tier becomes comfortable for normal use.

Decision

Choose Your Path

PATH	EFFORT	OUTCOME
Optimize architecture	1–2 hrs dev	Free tier sufficient
Enable Gemini billing	2 min	Limits gone · ~RM 5/month
Both (recommended)	2 hrs + 2 min	Cheapest + most reliable

Recommendation: **Both**.
Optimize once for cleaner UX. Billing as the safety net.